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**Megger (India) Pvt Limited**

211, Crystal Paradise Mall,

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T + 91 22 26740468 F + 91 22 26740465

E Nupoor.Charde@megger.com

W www.megger.com

metrology lab?

The majority of metrology labs are good and effective. Unfortunately, there are exceptions. Some companies look for the lab that is the lowest bidder without studying other qualifications. It is up to engineers to verify the veracity of the lab. A great lab will be happy to show you around and will be proud of the traceability of its standards. Take the calibration manual for your instrument and sit with the lab personnel. Verify that the lab has the necessary instruments to perform accurate calibration.

You must also ensure that a lab's instruments are properly calibrated with traceability. Most countries have their own national standards laboratory for this; in the US, it is the National Institute of Standards and Technology (NIST). A good metrology lab has records showing that its instruments compare correctly to a chain of standards that go back to a master standard maintained by the NIST or other national standard.

Internal components of test instruments (such as voltage references, input dividers, amplifier gain and offset) can drift over time. A good calibration schedule ensures that the typically minor drift does not impair the measurement. It is this drifting that calibration finds and corrects. But accidents do happen; instruments can be dropped or one can slip and probe a high voltage. For example, a digital multimeter (DMM) can be overloaded, causing a large error. Some mistakenly think that this does not produce an out of calibration error as the inputs are fused or breaker protected. However, a high voltage can jump across the input-protection device, or a transient can destroy the circuit before the protective device has time to react.

When we send an instrument for calibration, we expect the metrology lab to bring the instrument back into calibration. Engineers should also receive a report showing how far out of calibration the instrument was before, and how far it is after adjustment. If the report shows significant calibration errors, it may be necessary to redo a project that was completed with the

instrument just prior to calibration and take new measurements.

How often is calibration required?

There is no one answer to this query, because instruments, environments and applications vary. Test instrument manufacturers recommend calibration intervals for typical conditions; extreme conditions and very critical measurements may require more frequent calibration. Here are some general categories of calibration intervals:

Routine calibration as required by customer contract, quality standard organisation, military specifications, or other industry requirements. It is necessary to review the applicable requirements before the test to ensure that the calibration or certification for the test equipment is met.

Before and after a key measuring project. For example, when a new product pilot run is complete, a design engineer will characterise the product to ensure that it meets specifications and will optimise the test procedure. Final test adjustments made here can substantially decrease test time and affect profitability. Complete and reliable testing requires that the state of the instruments be verified both before and after the test period.

When you suspect a measurement is faulty or when the instrument has been overloaded or dropped. It is important to check the calibration and safety fidelity (such as a wire shorting to the case if it was dropped).

Accurate calibration is not a needless luxury—it ensures the reliability of test instruments and even the safety of personnel. For example, before working on a piece of equipment, one might measure a meter's voltage to be sure it is safe. If a meter is broken or gives inaccurate information, it could result in injury or death. Additionally, calibration assures quality. It guarantees the accurate test results necessary to verify that products in final test and those shipped to customers indeed meet specifications. ●

The author is a strategic applications engineer at Maxim Integrated